

Module 3 — Read 0

Project preview — Project 3 — Discovering Patterns

Module 3, Read 0: Why exploratory analysis matters

Big question: What does a dataset *look like*, and what patterns hide inside it?

Project: [Project 3 — Discovering Patterns](#) · **Sample report:** [TidyTuesday EDA example](#) · **Next:** [Read 1](#)

i Short read — start here

Five-minute orientation before [Read 1](#). Modules 1 and 2 focused on a *single* variable — testing a claim or estimating one proportion/mean. Module 3 asks “*What patterns live across several variables in this dataset?*”

The story in one pass

In Modules 1–2 you had **one** variable at a time: a proportion to test, or a mean to estimate. In Module 3 you explore a **real dataset** with multiple variables by making **graphs** and computing **numeric summaries** for each variable — and then for pairs of variables together.

Exploratory data analysis (EDA) is not about hypothesis testing. It is about building an honest picture of what the data contain before drawing conclusions. The four key tools for Project 3:

Variable types	Graph	Numeric summary
One numerical	Histogram or dot plot	Mean, median, SD, IQR
One categorical	Bar chart	Count, proportion
Numerical by group	Boxplot per group	<code>group_by() + summarize()</code>
Two numerical	Scatterplot	Form, direction, strength
Two categorical	Stacked bar chart	Two-way table (row/column proportions)

Project 3 is **descriptive only** — inferential tests (t-test, etc.) are optional enrichment, not required.

What makes a good Project 3 dataset

A good dataset for Project 3 has:

- **4 variables** of different types (at least one numerical, at least one categorical)
- **10 rows** of observational units (more is fine)
- A **documented source** (TidyTuesday, government open data, UCI ML repository, etc.)
- Variables with a real-world story worth telling

Tour what you will do

Project step	What it means
Choose a dataset	Pick from TidyTuesday or another approved source. Document the URL.
Classify four variables	For each: numerical (continuous or discrete) or categorical (and binary if yes/no).

Project step	What it means
Univariate summaries	One graph and one numeric fact per variable (4 variables).
Bivariate exploration	Choose at least 2 variable pairs; use the right graph/table for each type combination.
One-page write-up	Two best graphics, 4 labeled numeric facts, four findings in plain language.

What to do next

1. Glance at [Project 3](#) and note due dates on the [course path](#).
2. Browse a few [TidyTuesday datasets](#) — pick one that interests you.
3. Start **Read 1** — comparing a numerical variable across two groups with boxplots and `group_by()`.

When you finish Read 1, take [Quiz A \(Def & Notation\)](#), then continue with [Read 2](#).